HW6

Hyeonseo Daniel Kim

Plot the x-t diagram for a sudden expansion wave. Take a tube whose initial length is given as 5 meters filled with air at 101325 Pa and 300 K. At time t = 0 one of the endcaps moves away from the other endcap at a constant speed of 100 m/s. Plot graphically the characteristics associated with the expansion fan as it emanates from from the sudden expansion corner in the x-t diagram and subsequently interacts with its reflection coming from the second stationary endcap

Part 1 : Incident part of expansion Wave

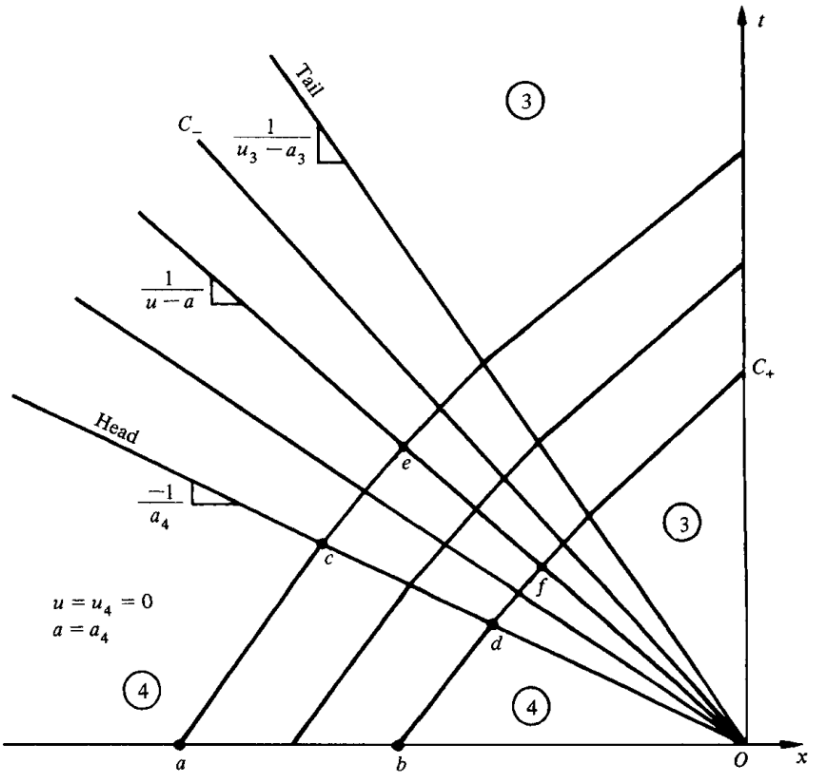


Figure 7.15 | The C_+ and C_- characteristics for a centered expansion wave (on an xt diagram).

In the image,
$(J_+)_a = (J_+)_b$
$(J_+)_a = (J_+)_c = (J_+)_e$
$(J_+)_b = (J_+)_d = (J_+)_f$

where

$$J_+ = u + \frac{2a}{\gamma - 1} = Const$$

$$J_- = u - \frac{2a}{\gamma - 1} = Const$$

In the given problem, for the head,

u = u_4 = 0

u_p = u_3 = 100

linspacing u_p from 100 to 0 by given n increments...

In [1]:

```
import numpy as np
import matplotlib.pyplot as plt

#Problem given
gamma = 1.4
R= 287
P_init = 101325
T_init  = 300
u_p = 100

#Number of rays = 50
n_rays = 4

# 0 to piston speed, divide into n rays
u = np.linspace(0,u_p, n_rays)
u_4 = 0

x_init, y_init = 5.0, 0.0
x_wall = 0.0
x_linspace = np.linspace(0,5,100)

incident = []
incident_wall = []

a_4 = np.sqrt( gamma * R * T_init)
print(a_4)

fig,ax = plt.subplots()

def incidentray(u, a):
    m = 1 / (u - a)
    y_wall = y_init + m * (x_wall - x_init)
    return y_wall, m

#Jplus hardcoded to be identical across incident rays, J minus changes
J_plus_incident = u_4 + (2 * a_4)/(gamma -1)
for i in u:

    a_i = ((J_plus_incident - i)*(gamma - 1))/2
    J_minus_incident = i - (2*a_i)/(gamma -1)
    _, m_i = incidentray(u,a_i)
    incident.append([i, a_i, m_i, J_plus_incident, J_minus_incident])
```

```
for idx, (u, a, m_i, J_plus, J_minus) in enumerate(incident):
    y_wall_i, _ = incidentray(u,a)
    ax.plot([x_init, x_wall], [y_init, y_wall_i ])
    J_plus_wall = ((2*a)/(gamma - 1)) # + u_wall = 0
    incident_wall.append([idx, y_wall_i, u, a, J_plus_wall, J_minus])

print(incident_wall)
for idx, y_wall_i, u, a, J_plus_wall, J_minus in incident_wall:
    ax.plot(x_wall, y_wall_i, '.', color = "black")

#fig.show()

# 5, 8 , 10 has direction = negative
# Others have direction = Positive
def correctvalues(Jplus, Jminus):
    u = 0.5 * (Jplus + Jminus)
    a = (gamma - 1) / 4 * (Jplus - Jminus)
    return u, a

def marching(direction, u1, a1, u2, a2):
    if direction == "right":
        theta1 = np.arctan2(1, u1 + a1)
        theta2 = np.arctan2(1, u2 + a2)
    elif direction == "left":
        theta1 = np.arctan2(1, u1 - a1)
        theta2 = np.arctan2(1, u2 - a2)
    m_avg = np.tan(0.5 * (theta1 + theta2))
    return m_avg

#    incident_wall.append([idx, y_wall_i, u, a, J_plus_wall, J_minus])

J_plus  = incident_wall[0][4]   # idx = 1
J_minus = incident_wall[1][5]   # idx = 2
print(correctvalues(J_plus, J_minus))
#This value is same as the 2 value above, since there hasn't been any reflection mo

u_1, a_1 = incident_wall[0][2], incident_wall[0][3]
u_2, a_2 = correctvalues(J_plus,J_minus)

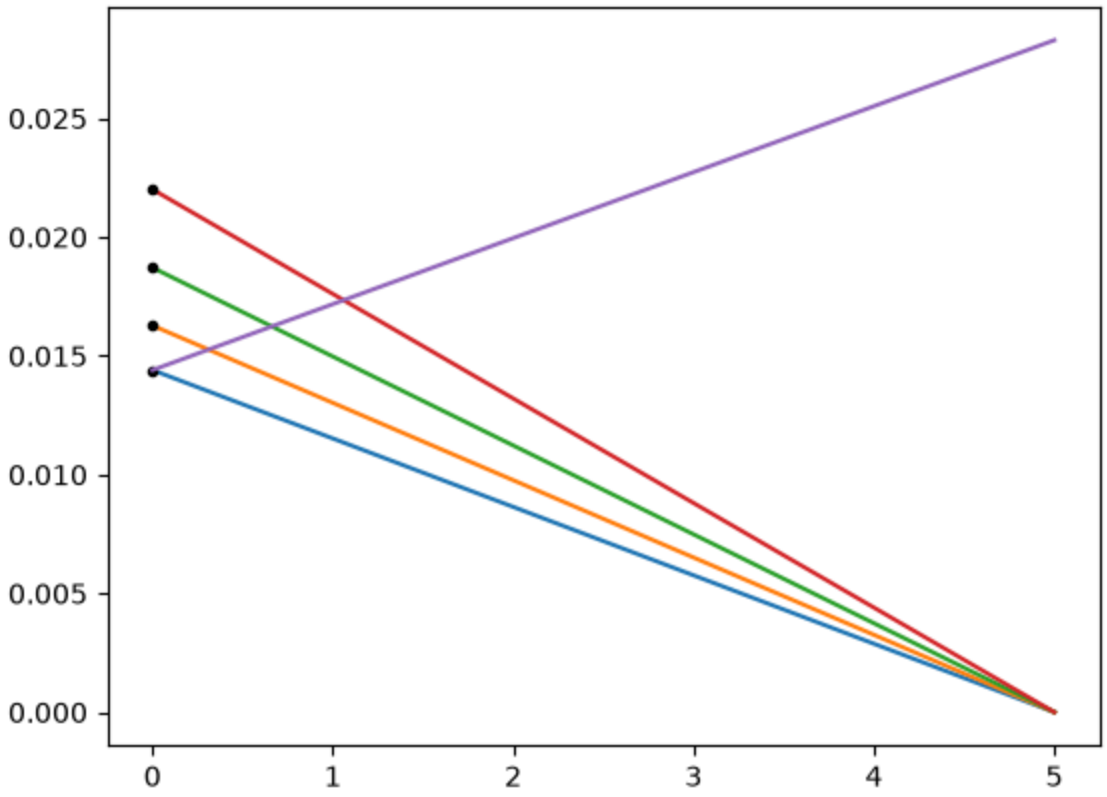
m_12 = marching("right", u_1, a_1, u_2, a_2)
print(m_12)

y_12 = incident_wall[0][1] + m_12*x_linspace

ax.plot(x_linspace,y_12)

plt.show()
```

```
347.18870949384285
[[0, np.float64(0.014401389974026995), np.float64(0.0), np.float64(347.1887094938428
5), np.float64(1735.9435474692145), np.float64(-1735.9435474692145)], [1, np.float64
(0.016276639881194), np.float64(33.333333333333336), np.float64(340.52204282717616),
np.float64(1702.6102141358813), np.float64(-1669.276880802548)], [2, np.float64(0.01
8713365581472005), np.float64(66.66666666666667), np.float64(333.8553761605095), np.
float64(1669.2768808025478), np.float64(-1602.610214135881)], [3, np.float64(0.02200
81359286717), np.float64(100.0), np.float64(327.18870949384285), np.float64(1635.943
5474692145), np.float64(-1535.9435474692145)]]
(np.float64(33.33333333333326), np.float64(340.52204282717616))
0.0027775545354049506
```



Significant part of the data in each index of incident_wall is wrong.

all data for the first point is correct.
All J minus of the curve is correct.
At the current wall points, u = 0.
I can update the u values

I can find correct a with J_minus
I can update the index with the correct a
Based on correct u and a, I can find correct J_plus numbers
I can update the index with the correct J_plus numbers

For point 2, there hasn't been interaction yet.
slope can be calculated with J+ and J- value as-is

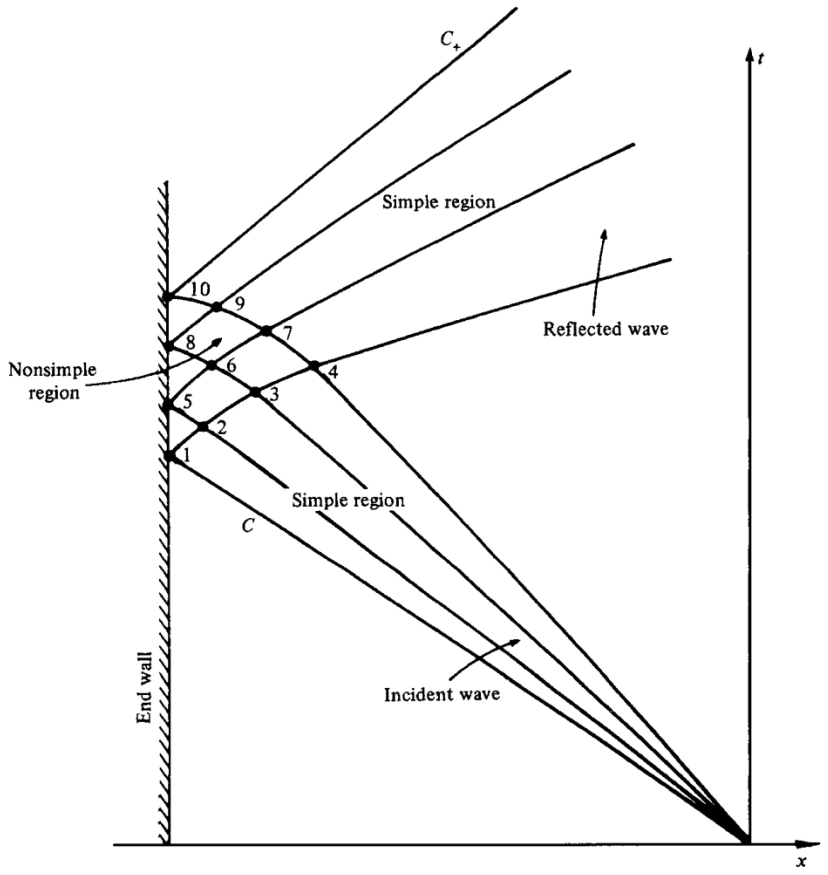


Figure 7.16 | Reflected expansion wave on an xt diagram.

Eqs. (7.73) through (7.76). For example, J_+ and J_- at points 1, 2, 3, and 4 are known from the incident expansion wave. At point 5, a_5 is determined by $(J_-)_5 = (J_-)_2$ and by the boundary condition $u_5 = 0$. At point 6, both a_6 and u_6 are determined from Eqs. (7.75) and (7.76), knowing that $(J_-)_6 = (J_-)_3$ and $(J_+)_6 = (J_+)_5$. The location of point 6 in the xt space is found by the intersection of the C_- characteristic through point 3 and the C_+ characteristic through point 5. These characteristics are drawn as straight lines with slopes that are averages between the connecting points. For example, for line 3–6,

$$\tan^{-1}\left(\frac{dt}{dx}\right) = \frac{1}{2}\left[\tan^{-1}\left(\frac{1}{u-a}\right)_3 + \tan^{-1}\left(\frac{1}{u-a}\right)_6\right]$$

and for line 5–6,

$$\tan^{-1}\left(\frac{dt}{dx}\right) = \frac{1}{2}\left[\tan^{-1}\left(\frac{1}{u+a}\right)_5 + \tan^{-1}\left(\frac{1}{u+a}\right)_6\right]$$